

Shengyun Si

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Date of Birth: 14.10.1999 | Nationality: Chinese

EDUCATION

Technical University of Munich <i>PhD Student - advised by Prof. Michael Backes</i>	Munich, Germany 2026.03-now
Technical University of Munich <i>Master of Robotics, Cognition, Intelligence</i>	Munich, Germany 2021.10-2025.02
Zhejiang University <i>Bachelor of Automotive Engineering</i>	Hangzhou, China 2017.09-2021.7

RESEARCH AND INDUSTRY INTEREST

My long-term research goal is to develop a system capable of grounding and reasoning over world knowledge to enable safe and meaningful interactions with humans. Motivated by this goal, I am mainly interested in **LLM-based Multiagent System**, **Large Language Models Safety**, and **Multimodal Language Model Applications**.

PUBLICATION

Shengyun S., Xinpeng W., Guangyao Z., Nassir N, Barbara P.
Think Before Refusal: Triggering Safety Reflection in LLMs to Mitigate False Refusal Behavior. [under review]
[paper]

Shengyun S.*, Haowei Z.*, Yilun Z., Linyong N, Xiangru T, Arman C.
OpenT2T: An Open-Source Toolkit for Table-to-Text Generation. EMNLP 2024
[paper]

Shengyun S.*, Yilun Z.*, Haowei Z.*, Linyong N, Xiangru T, Arman C.
Investigating Table-to-Text Generation Capabilities of Large Language Models in Real-World Information Seeking Scenarios. EMNLP 2023
[paper]

Shuo C., Zhen H., Haokun C., Bailan H., **Shengyun S.**, Jingpei W., Philip T., Volker T., Jindong G.
Bag of Tricks for Subverting Reasoning-based Safety Guardrails. Arxiv Preprint
[paper]

EXPERIENCE

PhD Researcher <i>CISPA Helmholtz Center for Information Security</i> Conduct scientific research on LLM safety and other related topics	Mar 2026 - Now Saarbrücken, Germany
Research Assistant <i>DFKI (German Research Centre for Artificial Intelligence)</i> - Developed and evaluated steering vector methods for controlling the behavior of MLLMs, enabling fine-grained manipulation of model outputs such as style, reasoning depth, and factuality. - Explored training-free representation engineering approaches to improve the reliability and safety of MLLMs, particularly in handling ambiguous or conflicting multimodal inputs. - Documented research findings and contributed to internal reports, laying groundwork for potential publications in top-tier NLP and multimodal AI venues.	June 2025 - Feb 2026 Berlin, Germany
Data Scientist Intern <i>Allianz SE</i> Design and implement RAG-based document answering pipelines using Haystack and LLM, aiding finance experts in streamlining their workflow.	October 2023 - March 2025 Munich, Germany

- Developed a backend API using **FastAPI** and **Haystack** to integrate prototypes into a production environment, facilitating interactions with the frontend and vector databases. Managed the deployment of necessary services using **Kubernetes**
- Fine-tuned the **sentenceTransformer** based ranker and retriever models in the pipeline using a collected ground truth dataset to achieve higher accuracy. After fine-tuning, the models' recall on the validation set improved from **72%** to **90%**.
- Developed a command-line tool using **Poetry** and **Click** libraries to simplify the workflow for non-technical team members, enabling them to independently manage and execute business logic.
- Collaborate with a multidisciplinary team to understand financial domain-specific challenges, ensuring the developed tools were aptly tailored to user needs.
- Authored technical documentation to assist non-technical colleagues in understanding and utilizing command-line tools for tasks such as one-click training and deployment.

Research Assistant

March 2023 - Sep 2023

Yale University LILY Lab

Munich, Germany(remote)

- Research on how **LLMs**, like **GPT3.5/GPT4/LLaMa** perform on **Table2Text** task. If LLMs show great performance in this task, we further research on how fine-tuned model benefit from LLMs' strong table-to-text abilities.
- Fine-tune models with different table-to-text datasets **LogicNLG/WTQ/Totto/FetaQA**. Evaluate the faithfulness of generation based on different metrics. Prompt **LLMs** to generate faithful statements and give feedback to previous fine-tuned models' generation with CoT.
- Demonstrate **LLMs** can generate statements with higher faithfulness compared with previous state-of-the-art fine-tuned models. We also demonstrate that **LLMs** using chain-of-thought prompting can generate high-fidelity natural language feedback for other table-to-text models' generations, provide insights for future work regarding the distillation of text generation capabilities from **LLMs** to smaller models.
- Propose **OpenT2T**, the first open-source toolkit for table-to-text generation, designed to reproduce existing table pre-training models for performance comparison and expedite the development of new models. We have implemented and compared 7 fine-tuned models as well as 44 large language models under zero- and few-shot settings on 9 table-to-text generation datasets, covering data insight generation, table summarization, and free-form table question answering
- Our work has been accepted by the **EMNLP2023, EMNLP2024**.
- <https://aclanthology.org/2023.emnlp-industry.17/>
- <https://aclanthology.org/2024.emnlp-demo.27/>

NLP Research Intern

June 2023 - Sep 2023

Siemens AG

Munich, Germany

- Utilizing the latest methods in natural language processing, review scholarly articles and propose recommendations for practical implementation.
- Fine-tune pre-trained models on industrial data, for example the Requirement-Model alignment data which is generated during the traces of industrial production. Use the fine-tuned model to serve for the upcoming raw data.

SELECTED AWARDS

- **2021:** Outstanding Graduates at Zhejiang University
- **2019:** Second-grade Scholarship at Zhejiang University
- **2018:** National Encouragement Scholarship at Zhejiang University

TECHNICAL SKILLS

Languages: English(business-fluent), German(B2), Chinese(Native)

Frameworks: Tensorflow, Pytorch, NumPy, Scikit-learn, Matplotlib, Docker

Big data technologies: Hadoop, Spark, Dask, MLlib, Apache OpenNLP

Programming Language: Python, C++/C, Java, Scala